COURSEWORK 1: BEAT TRACKING

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1. INTRODUCTION

A beat tracker is implemented based on the work of Ellis [1], which combines a step of tempo induction with a dynamic programming algorithm. The solution is designed to work on the Ballroom dataset, and therefore was adjusted using a train subset and tested in a test subset of the same dataset. In addition, a basic downbeat detector is proposed and tested.

2. IMPLEMENTATION

A beat tracker is implemented following the algorithm described by Ellis in [1]. This choice was made judging the simplicity of the algorithm and considering the Ballroom dataset contains tracks with little variation of bpm over time.

The detection consists of two stages. First, tempo induction is conducted to obtain the expected period between beats. Second, this expected period is used in the context of a dynamic programming algorithm to detect beats by maximising the score function defined in Equation (1).

2.1 Dynamic Programming formulation

Given a constant ideal interval of samples between beats τ_p, and a detected sequence of N beats t_i, the score function $C(t_i)$ is defined as

$$C(\{t_i\}) = \sum_{i=1}^{N} O(t_i) + \alpha \sum_{i=2}^{N} F(t_i - t_{i-1}, \tau_p) \quad (1)$$

where $O(t)$ is an onset strength envelope obtained by processing the audio, which is large at moments where there probably is a beat, $F(\tau_t, \tau_p)$ is a loss function that measures the consistency between the interval between detected beats, τ_t, and the ideal spacing between beats, τ_p, and α balances the two terms.

The loss function $F(\tau_t, \tau_p)$ is proposed to be

$$F(\tau_t, \tau_p) = -\left(\log \frac{\tau_t}{\tau_p}\right)^2 \quad (2)$$

which takes a value of 0 when the interval between beats τ_t matches the ideal interval τ_p, and decreases otherwise.

The score for the best-scoring sequence of beats t_i that ends at time t can be written as

$$C^*(t) = O(t) + \max_{\tau=0\ldots t} \{\alpha F(t-\tau, \tau_p) + C^*(\tau)\} \quad (3)$$

The previous beats can be found recursively, as the preceding beat with the best score is

$$P^*(t) = \arg\max_{\tau=0\ldots t} \{\alpha F(t-\tau, \tau_p) + C^*(\tau)\} \quad (4)$$

2.2 Onset strength envelope

The steps for obtaining the onset strength envelope of a signal are the following:

- Resample the signal to 8kHz.
- Calculate the short-term Fourier Transform (STFT) magnitude, using time windows of 32 ms, and a hop of 4 ms.
- Map the STFT to a Mel spectrogram of 40 Mel bands, and convert it to dB.
- Calculate the first-order difference in time.
- Set negative values of the difference to zero.
- For each time window, sum across frequency bands to obtain a single value.
- Filter the signal with a high-pass filter with a cutoff of 0.4Hz to make it locally zero mean.
- Smooth the signal by convolving with a Gaussian envelope of 20ms.
- Normalise the signal by dividing it by its standard deviation.

2.3 Tempo induction

2.3.1 Basic tempo induction

The first algorithm for finding a global tempo of a track consists of finding the value of the interval between beats, τ, which maximises the tempo period strength defined as

$$TPS(\tau) = W(\tau) \sum_t O(t) O(t-\tau) \quad (5)$$

where $O(t)$ is the onset strength envelope described in section 2.2, and $W(\tau)$ is a Gaussian weighting function that is used to favour tempi that are closer to a certain predefined tempo, accounting for documented perception bias, as defined in equation 6. This perceptual bias is documented to be around 120 BPM, and therefore, the corresponding interval of samples for this value is set as τ_0. σ_τ,

on the other hand, is set at 1.4 octaves, as stated in [1], which was the best value found in their experiments.

$$W(\tau) = \exp\left\{-\frac{1}{2}\left(\frac{\log_2 \tau/\tau_0}{\sigma_\tau}\right)^2\right\} \quad (6)$$

It is worth noting that since the train set tracks' tempi have values between 70 BPM and 225 BPM approximately, this condition was enforced by zeroing the TPS function for frames outside of that region before searching for the maximum value.

2.3.2 Refined tempo induction, considering double and triple meters

A more refined version of the algorithm is proposed in [1], by defining two auxiliary tempo period strength functions, one constructed to incorporate the value of TPS at 2τ (equation 7), and another one to incorporate it at 3τ (equation 8). It is worth noting that in this case the reported best value for σ_τ is 0.9 octaves.

$$TPS2(\tau) = TPS(\tau) + 0.5TPS(2\tau) + 0.25TPS(2\tau - 1) + 0.25TPS(2\tau + 1) \quad (7)$$

$$TPS3(\tau) = TPS(\tau) + 0.33TPS(3\tau) + 0.33TPS(3\tau - 1) + 0.33TPS(3\tau + 1) \quad (8)$$

The value of τ_{max} that maximises each expression is searched for, and the value of $TPS2$ and $TPS3$ at their respective maximums is compared. If the maximum value for $TPS2$ is greater than the one of $TPS3$, it is likely that the audio presents a binary subdivision of a bar, while the contrary would indicate a ternary subdivision. The estimated tempo corresponds to τ_{max} for the function with the greatest maximum value. A visualisation of the different tempo period strength functions is provided in Figure 1.

2.4 Downbeat tracking

A basic idea for downbeat tracking is proposed based on the already available blocks of the system. First, the basic tempo induction algorithm is used to estimate the tempo of the piece. Second, the refined tempo induction algorithm is utilised to find information on the metre (is it duple or triple?). Third, the beat tracking algorithm is run, but with the expected period between beats adjusted to the downbeat metrical level. That is, if the metre is duple, the piece is probably in 4/4, and the expected period between beats should be a four times the estimated one in the first place. The same is done for the triple metre, but taking three times the estimated period.

3. EVALUATION

3.1 Tempo induction

The tempo induction algorithms described in section 2.3 were evaluated for different values of σ_τ in the 0.5 to 1.9 octaves range, comparing the obtained value of BPM with the ground truth value. The mean absolute error for each case can be seen in Figure 2. Due to these results, the value of σ_τ for the basic tempo induction algorithm is set to 0.9 octaves, while for the refined tempo induction algorithm it is set to 0.5 octaves, as they minimise the error metric.

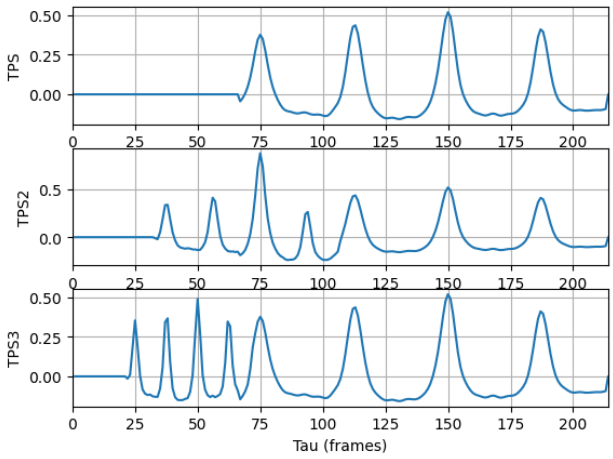


Figure 1. TPS, TPS2 and TPS3 for the samba track Albums-Macumba-05.wav. The track has a tempo of 100 BPM, which is detected correctly by the basic tempo induction approach. The refined tempo induction algorithm, on the other hand, indicates the tempo is 200 BPM and correctly detects the metre as duple.

In order to better understand where the algorithm is failing, the distribution of relative errors for tempo estimation is shown in Figure 3. It is clear that there are peaks in the relative error distributions in the values 0.5 and 1. A relative error of 0.5 implies the estimated BPM is half of the real BPM, while a relative error of 1 is encountered when the estimated BPM is double of the real BPM. For the basic tempo induction, the majority of errors seem to be due to estimating the BPM to be half of the expected BPM. In the case of refined tempo induction, the detected BPM tends to be the double of the expected BPM.

Given these results, it is decided to utilise the basic tempo induction algorithm, and reserve the refined tempo induction algorithm to obtain information about the bar subdivision.

3.2 Beat tracking

The optimal α value reported in [1] is 680. However, experiments were run on the train set to find the optimal value for this dataset. For a range of possible α values, the detected beats were compared to ground truth, and the F1 score is reported in Table 1. While further search could be conducted, 400 is considered as the optimal value of α.

4. RESULTS

The beat tracker was evaluated on the test subset, with the optimal configuration of parameters described in section 3. The obtained F1 scores for different styles and for different

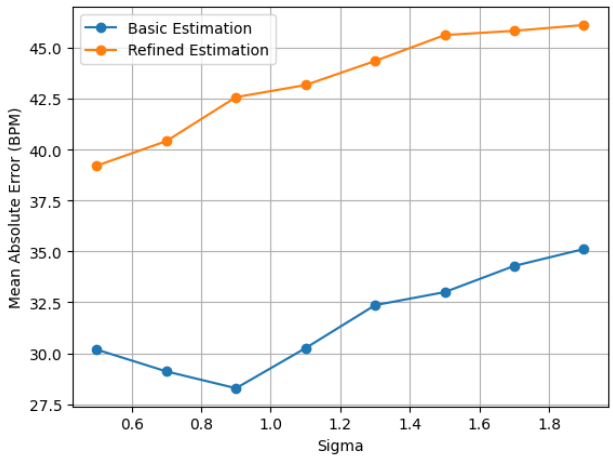


Figure 2. Mean absolute error in detected tempi for the basic and refined tempo induction algorithms in the train set.

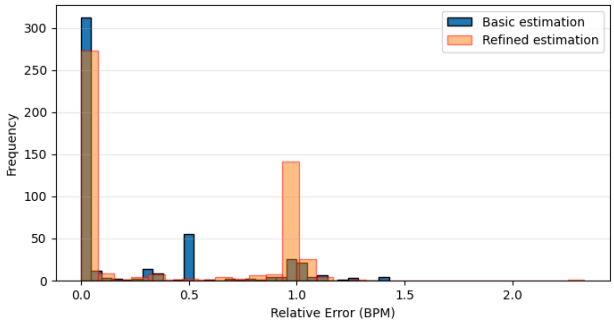


Figure 3. Distribution of relative error in detected BPM for the basic and refined tempo induction algorithms, each run with their optimal σ_τ value.

Table 1. F1 score for different values of α in train dataset

α	F1 Score
200	0.735
300	0.735
400	**0.736**
500	0.732
600	0.732
700	0.730
800	0.728
900	0.722
1000	0.723

metres are reported in Table 2. The beat tracker shows a better performance for tracks with a 4/4 metric structure, and for Rumba and ChaChaCha styles. An example of successful beat tracking is shown in Figure 4. Poor performance on Waltz should be further analysed, but octave errors in tempo induction could partially explain the results. The analysis of the failure case shown in Figure 5 shows another issue: while the tempo appears to have been correctly estimated, there is a phase issue in the detection, which is offset by half a period. A possible explanation for this case is that Samba tracks can be highly syncopated, presenting high energy at times that don't correspond to a beat. This can be seen in Figure 5 at around second 13. Additionally, once the first beat is chosen by the algorithm, there's no way of resetting its state, and the expected ideal period between beats could force the detection to continue with the offset.

Table 2. F1 Scores for Beat Detection by Dance Style and Metre in test dataset

Dance Style (n tracks)	**F1 Score**
Mean (210)	0.775
Rumba-American (1)	**1.000**
Rumba-International (12)	0.984
ChaChaCha (35)	0.980
VienneseWaltz (17)	0.875
Tango (29)	0.812
Samba (32)	0.755
Rumba-Misc (12)	0.751
Jive (15)	0.727
Quickstep (28)	0.697
Waltz (29)	0.467
Metre (n tracks)	**F1 Score**
Mean (210)	0.775
4/4 (164)	**0.857**
3/4 (46)	0.618

The downbeat tracker shows a poor result when evaluated on the test set, with a mean F1 score of 0.263. The mean F1 scores for different styles and metric structures are shown in Table 3. A successful downbeat detection example is shown in Figure 6. On the other hand, Figure 7 shows a failure case of downbeat detection, where the period between downbeats is correctly detected, but there is a phase offset: the second beat of each bar is detected as the downbeat, which is a reasonable error for this simple approach.

5. CONCLUSION

This work presented an implementation of the beat tracking system designed by Ellis in [1], with the addition of a downbeat tracking approximation. The algorithm was tuned to optimal parameters with the train dataset, defining the value of α as 400, and σ_τ as 0.9 octaves. The mean F1 score of the detection when evaluating on the test dataset is 0.775, and the system shows better results for dance styles such as Rumba and ChaChaCha, as well as for 4/4 metric

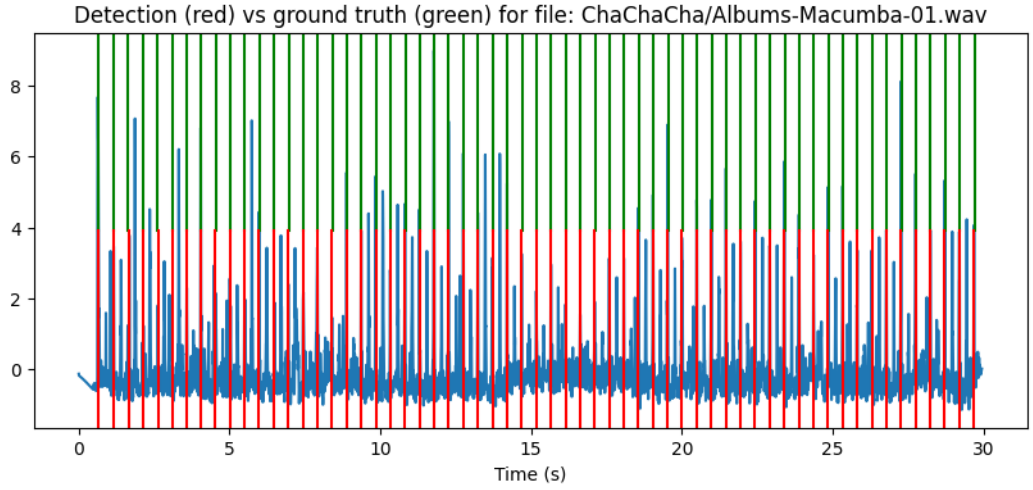


Figure 4. Analysis of a beat tracking success case.

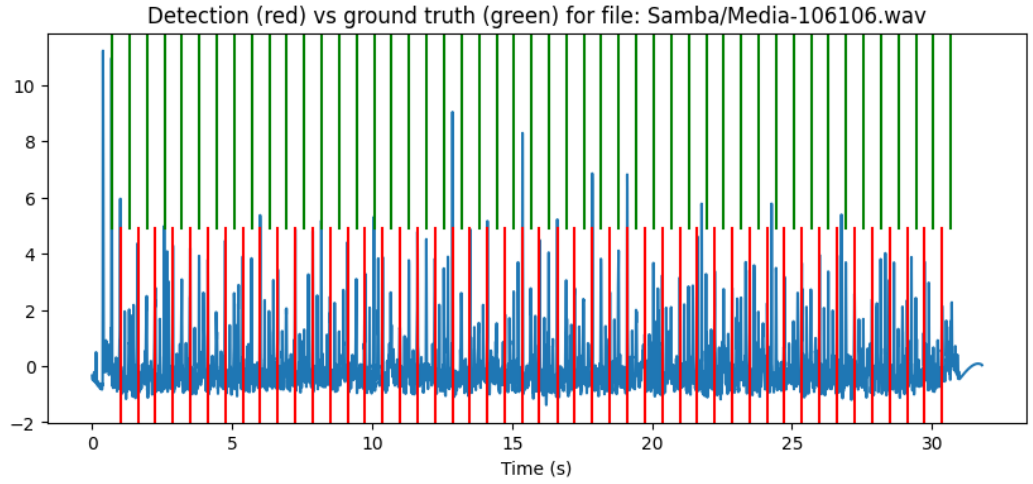


Figure 5. Analysis of a beat tracking failure case.

structures. The downbeat detection shows a poor result, with a mean F1 score of 0.263. The analysis shows errors related to incorrect tempo induction (octave errors), as well as phase offsets in the detected beats and downbeats.

Table 3. Downbeat F1 Scores by Dance Style and Metre in test dataset

Dance Style (n tracks)	**F1 Score**
Mean (210)	0.263
Rumba-International (12)	**0.397**
Samba (32)	0.372
Tango (29)	0.327
Quickstep (28)	0.280
VienneseWaltz (17)	0.262
Waltz (29)	0.232
ChaChaCha (35)	0.222
Rumba-Misc (12)	0.140
Jive (15)	0.038
Rumba-American (1)	0.000
Metre (n tracks)	**F1 Score**
Mean (210)	0.263
4/4 (164)	**0.268**
3/4 (46)	0.243

6. REFERENCES

[1] D. P. Ellis, "Beat tracking by dynamic programming," *Journal of New Music Research*, vol. 36, no. 1, pp. 51–60, 2007.

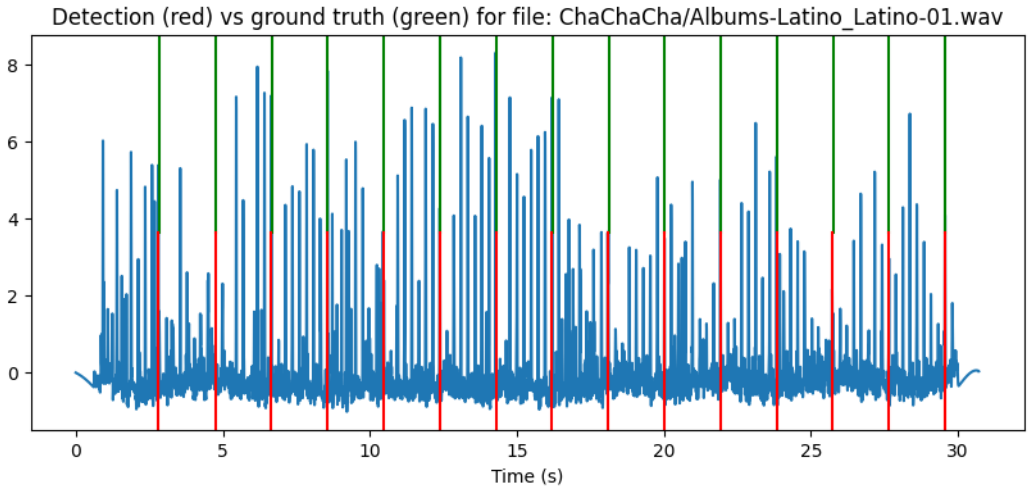


Figure 6. Analysis of a downbeat tracking success case.

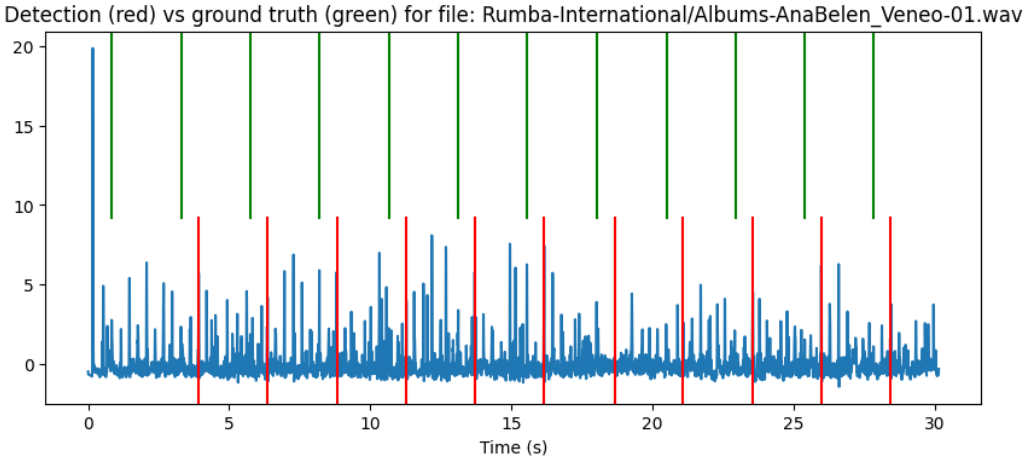


Figure 7. Analysis of a downbeat tracking failure case.